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3rd Sem / Civil, Brick Tech, Const Mgmt, Highway Engg.
Subject : Fluid Mechanics

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Unit of surface tension is (CO1)

- a) N/m b) N/m^2
- c) No unit d) N

Q.2 If M and G are at same point, the body is in (CO2)

- a) Stable equilibrium
- b) Unstable equilibrium
- c) Neutral equilibrium
- d) None

Q.3 Venturimeter is always placed in (CO3)

- a) Horizontal position b) Vertical position
- c) Inclined position d) Any of the above

Q.4 Which of the following is major energy loss? (CO4)

- a) Due to bend in pipe
- b) Due to sudden contraction in pipe
- c) Due to friction
- d) Due to pipe fitting

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Q.5 Which is dimensionless? (CO4)

- a) Reynold's number b) Mach's number
c) Euler's number d) All of these

Q.6 Rectangular channel section is most economical when (CO5)

- a) $d=b$ b) $d=b/2$
c) $d=2b$ d) $b/4$

Where d -depth of channel, b -Width of channel

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Vacuum pressure is also called _____ (CO1)

Q.8 The point where the resultant pressure acts is called _____ (CO2)

Q.9 Flow of fluid through a tapered pipe is _____ flow. (CO3)

Q.10 Dimensionless formula of velocity is _____ (CO5)

Q.11 The water in the canal flows under _____ (CO5)

Q.12 Priming is necessary for a _____ pump. (CO5)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 If specific gravity of fluid is 1.62, calculate mass density, specific weight and specific volume. (CO1)
- Q.14 Write a short note on differential manometer. (CO1)
- Q.15 State any two use of hydraulic coupling. (CO2)
- Q.16 Define the center of buoyancy, meta-center and meta-centric height. (CO1)
- Q.17 Write down any two limitations of Bernoulli's theorem. (CO3)
- Q.18 Discharge is measured by a rectangular notch of crest width equal to 0.3 m of a rectangular channel. If the depth of water above crest is 0.12 m and $C_d = 0.62$, find the discharge. (CO3)
- Q.19 Oil, having specific gravity 0.9 is flowing through a pipe of 0.15 m diameter at the rate of $0.0275 \text{ m}^3/\text{s}$. Find the type of flow if the viscosity of oil is 3.8 poise. (CO4)
- Q.20 Write short note on limitations of dimensionless analysis?
- Q.21 Enlist any five difference between flow through Open channel and Flow through pipe.

Q.22 Define the following : (CO5)

- a) Hydraulic pump
- b) Cavitation
- c) Priming
- d) Pitting

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 A rectangular block is 4m long, 3 m wide and 1.40 m high. The depth of immersion of the block is 1.0 m in sea water. If the center of gravity is 0.70 m above the bottom of the block, determine the metacentric height. Take the density of sea water is 1030 Kg/m^3 (CO2)

Q.24 A 30 cm diameter pipe, conveying water, branches into two pipes of diameter 20 cm and 15 cm respectively. If the average velocity in the 30 cm diameter pipe is 2.5 m/s, find the discharge in the pipe. Also determine the velocity in 15 cm pipe, if the average velocity in 20 cm pipe is 2 m/s. (CO3)

Q.25 Explain the construction and working of centrifugal pump with neat sketch. (CO5)